10.4 Cross-Sections

A **cross-section** is the 2D shape formed when a plane slices through a 3D solid. The shape you get depends entirely on the *orientation* of the cutting plane: parallel to a base, through an axis, through an apex, tilted, or through several faces at once. Beyond identifying the shape, this worksheet computes actual cross-section *dimensions and areas* — for a cone or pyramid cut parallel to the base, the cross-section is **similar** to the base, scaled by the ratio of distances from the apex (a direct application of similar triangles); for a sphere cut off-center, the Pythagorean Theorem relates the sphere's radius, the distance to the center, and the smaller circle's radius.

Examples

Example 1: Cylinder sliced parallel to its base

A cylinder is sliced by a plane parallel to its circular bases. Describe and justify the cross-section.

Solution

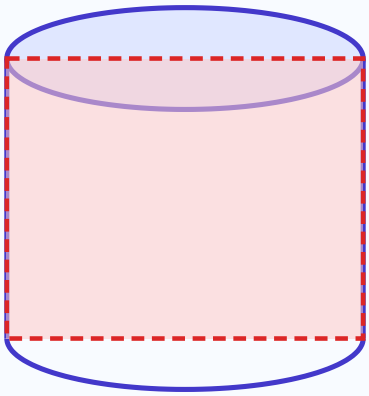
Every point on the cylinder's curved surface is the same distance r from the central axis. A plane parallel to the bases intersects the curved surface in a set of points all still that same distance r from the axis — that is a **circle** of radius r, congruent to the base.

Answer: a circle congruent to the base.

Example 2: Cylinder sliced through its axis

A cylinder has radius 4 and height 9. It is sliced by a plane through its central axis, perpendicular to the bases. Describe the cross-section and find its area.

Solution



The axial cross-section of a cylinder is a rectangle: width = diameter, height = cylinder's height.

The cut passes through the full width of the circle (a diameter, $2r$) and runs the full height h — this is a **rectangle** with dimensions $2r \times h$.

$2r = 2(4) = 8$; height $= 9$.

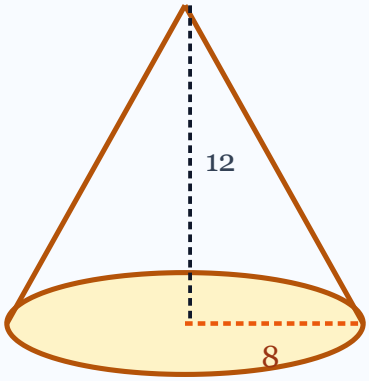
Area $= 8 \times 9 = 72$.

Answer: rectangle, 8×9, area 72 square units.

Example 3: Cone sliced parallel to its base — similar triangles

A cone has base radius 8 and height 12. A plane parallel to the base cuts the cone at a point 9 units from the apex (measured along the axis). Find the radius of the circular cross-section and its area.

Solution



Slicing parallel to the base at distance d from the apex gives a circle scaled by $\frac{d}{H}$.

The smaller cone (apex to cut) is similar to the whole cone, scaled by $\frac{d}{H} = \frac{9}{12} = \frac{3}{4}$.

Cross-section radius: $r' = R \cdot \frac{d}{H} = 8\,(\frac{3}{4}) = 6.$

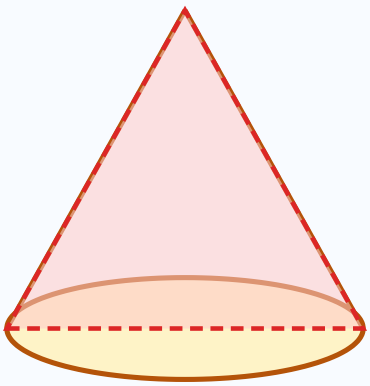
Area $= \pi r'^2 = \pi(36) = 36\pi.$

Answer: a circle of radius 6; area 36π.

Example 4: The *same* cone sliced through the apex and axis

Using the same cone (radius 8, height 12) from Example 3, now slice through the apex and the axis instead. Describe the cross-section and find its area.

Solution



The axial cross-section of a cone is an isosceles triangle: base = diameter, height = the cone's height.

This cut produces an **isosceles triangle** with base $= 2R = 16$ and height $= H = 12$ (the two equal sides run from the apex to each end of the base diameter).

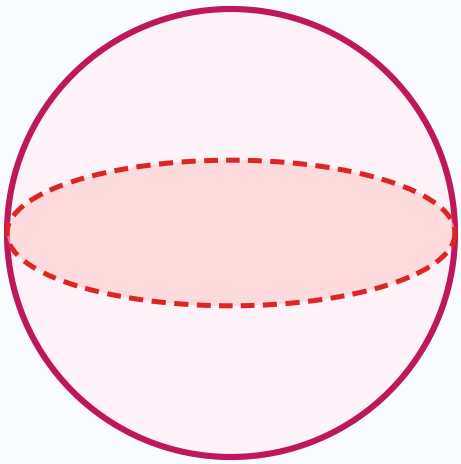
$$\text{Area} = \frac{1}{2}(\text{base})(\text{height}) = \frac{1}{2}(16)(12) = 96.$$

Answer: isosceles triangle, base 16, height 12, area 96 square units. *Compare to Example 3 — the same cone gives very different cross-sections depending on the cut.*

Example 5: Sphere sliced through its center

A sphere has radius 7. It is sliced by a plane through its center. Describe the cross-section and find its area.

Solution



A slice through the center is the largest possible circular cross-section: a great circle.

A cut through the center produces the largest possible circle on the sphere, called a **great circle**, with the same radius as the sphere: $r = 7$.

Area $= \pi r^2 = 49\pi$.

Answer: a great circle of radius 7; area 49π.

Example 6: Sphere sliced off-center — Pythagorean Theorem

A sphere has radius 17. It is sliced by a plane 8 units from the center (not through the center). Find the radius of the circular cross-section and its area.

Solution

Let r be the cross-section's radius. The sphere's radius R, the distance d from center to plane, and r form a right triangle (a radius to the circle's edge, the perpendicular from center to plane, and the segment on the plane):

$R^2 = d^2 + r^2 \;\Rightarrow\; 17^2 = 8^2 + r^2 \;\Rightarrow\; 289 = 64 + r^2 \;\Rightarrow\; r^2 = 225 \;\Rightarrow\; r = 15.$

(This is the 8-15-17 Pythagorean triple.)

Area $= \pi(15)^2 = 225\pi.$

Answer: radius 15; area 225π — smaller than the great circle, as expected.

Example 7: Square pyramid sliced parallel to its base

A square pyramid has base edge 8 and height 12. A plane parallel to the base cuts it 9 units from the apex. Find the edge length of the cross-section and its area.

Solution

Just like the cone, the smaller pyramid (apex to cut) is similar to the whole pyramid, scaled by $\frac{9}{12} = \frac{3}{4}$.

Cross-section edge: $s' = 8\left(\frac{3}{4}\right) = 6.$

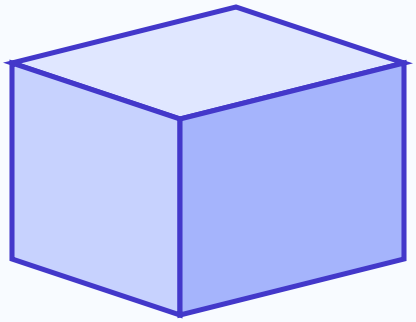
Area (a square) $= 6^2 = 36.$

Answer: a square of side 6; area 36 square units.

Example 8: Slicing off a cube's corner

A cube has edge length 6. A plane passes through the three vertices adjacent to one corner (cutting that corner off). Describe and compute the cross-section.

Solution



Cutting through the three vertices nearest a corner always produces an equilateral triangle.

Each side of the cut triangle connects two vertices that are a face-diagonal apart (they share a face of the cube), so each side has length $6\sqrt{2}$ — and all three sides are equal by the cube's symmetry, so the cross-section is **equilateral**.

Area of an equilateral triangle with side a: $\frac{\sqrt{3}}{4}a^2$, with $a = 6\sqrt{2}$, so $a^2 = 72$.

Area $= \frac{\sqrt{3}}{4}(72) = 18\sqrt{3}$.

Answer: equilateral triangle, side $6\sqrt{2}$, area $18\sqrt{3}$ square units.

Example 9: Slicing a cube through four vertices

A cube has edge length 5. A plane passes through four vertices, cutting straight through the middle (not through any corner). Describe and compute the cross-section.

Solution

This plane cuts through two opposite vertical edges' worth of faces, producing a **rectangle** whose sides are one edge of the cube (5) and one face diagonal ($5\sqrt{2}$).

Area $= 5 \times 5\sqrt{2} = 25\sqrt{2}$.

Answer: rectangle $5 \times 5\sqrt{2}$; area $25\sqrt{2}$ square units.

Example 10: Cylinder sliced by a tilted plane

A cylinder has radius 4. A plane tilted $60°$ from the horizontal (i.e., $60°$ from a plane perpendicular to the axis) slices through it without hitting either base. Describe the cross-section and estimate its dimensions.

Solution

A tilted cut (neither parallel nor perpendicular to the axis) produces an **ellipse**. One axis of the ellipse still equals the cylinder's diameter (the direction unaffected by the tilt) — the semi-minor axis is $r = 4$. The semi-major axis stretches by the tilt: $a = \dfrac{r}{\cos\theta} = \dfrac{4}{\cos 60°} = \dfrac{4}{0.5} = 8$.

Answer: an ellipse with semi-minor axis 4 and semi-major axis 8.

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Your Turn — Show Your Work!

For each problem, solve it and verify your solution. Show all steps clearly.

Problem 1: Cylinder parallel

Cylinder cut parallel to its base. Name the cross-section.

Answer: **Verification:**

Problem 2: Cylinder axial

Cylinder radius 4, height 9, cut through its axis. Find the cross-section's dimensions and area.

Answer: **Verification:**

Problem 3: Cone parallel

Cone radius 9, height 12, cut parallel to the base at 8 units from the apex. Find the cross-section's radius and area.

Answer: **Verification:**

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Problem 4: Cone axial

The same cone (radius 9, height 12) cut through the apex and axis. Find the cross-section's shape and area.

Answer: **Verification:**

Problem 5: Sphere center

Sphere radius 10, cut through the center. Find the cross-section's area.

Answer: **Verification:**

Problem 6: Sphere off-center

Sphere radius 17, cut 8 units from the center. Find the cross-section's radius and area.

Answer: **Verification:**

Problem 7: Pyramid parallel

Square pyramid, base edge 10, height 15, cut parallel to the base at 9 units from the apex. Find the cross-section's edge and area.

Answer: **Verification:**

Problem 8: Cube corner

Cube edge 8, corner sliced off through the three adjacent vertices. Find the cross-section's side length and area.

Answer: **Verification:**

Problem 9: Cube through four vertices

Cube edge 7, sliced through four vertices (not a corner). Find the cross-section's shape and area.

Answer: **Verification:**

Problem 10: Cylinder tilt

Cylinder radius 5, sliced by a plane tilted 45° from horizontal. Find the ellipse's semi-minor and semi-major axes.

Answer: **Verification:**

Problem 11: Prism parallel

Triangular prism cut parallel to its triangular base. Name the cross-section.

Answer: **Verification:**

Problem 12: Vocabulary

Define "great circle."

Answer: **Verification:**

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Problem 13: Concept

Under what condition does a cone's cross-section become an ellipse instead of a circle or triangle?

Answer: **Verification:**

Problem 14: Word problem

A cylindrical log of radius 6 in is cut by a plane parallel to its ends. Name the cross-section and find its area.

Answer: **Verification:**

Problem 15: Challenge

A cone has base radius 10 and height 24. A plane parallel to the base cuts the cone so the circular cross-section has area 25π. How far from the apex is the cutting plane?

Answer: **Verification:**

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✓ Answer Key

1. Circle (congruent to the base)
2. Rectangle 8×9; area 72
3. $r = 6$; area 36π
4. Isosceles triangle, base 18, height 12; area 108
5. 100π
6. $r = 15$; area 225π
7. edge 6; area 36
8. side $8\sqrt{2}$; area $32\sqrt{3}$
9. Rectangle $7 \times 7\sqrt{2}$; area $49\sqrt{2}$
10. semi-minor 5; semi-major $5\sqrt{2}$
11. Triangle congruent to the base
12. The circular cross-section of a sphere through its center (largest possible)
13. When the cutting plane is tilted (not parallel to the base and not through the apex/axis)
14. Circle; area 36π in^2
15. 12 units from the apex

Great work — keep practising!